

Scope 1

How I Handled Scope 1 of the Task

I started by focusing on a single field (Mood) to deeply understand the data structure: how Program (parent) and Season records are linked, and how counts, values, and status are represented in the sheet. Based on this understanding, I defined a clear goal: identify parent titles where the field is missing, and safely inherit values from seasons only when there is no conflict between seasons.

For the Mood sheet, I isolated all cases where the Program Mood is missing but at least one Season Mood is filled. For each Program Id, I checked the distinct Season Mood Values: if all seasons agreed on the same mood, I marked that program as **SAFE** and proposed inheriting that mood to the parent; if seasons disagreed, I marked it as **REVIEW**, meaning it requires manual validation and no automatic copy is applied.

After proving this logic on Mood, I generalized it so I could apply the same safe-inheritance rule to all other fields (Set_location, Dialect, Genre, Theme, Relation, Era, Plot). I created a small configuration structure listing each sheet name and its corresponding Program/Season count and value columns, then iterated over this config to run the same inheritance procedure field by field.

During this generalization, I encountered data type issues (mixed float/string ids and value fields), so I added a normalization helper that forces key columns to strings and count columns to numeric before running the logic. This ensured consistent groupby/merge behavior across sheets and prevented type errors.

For each field, I produced an updated review file containing the original Program/Season metadata plus two new columns: Inheritance_Status_<Field> (SAFE, REVIEW, or NONE) and Proposed_Program_<Field> for the SAFE cases. I also built a metrics table summarizing, per field, how many program–season pairs were inheritance candidates, how many programs could be safely auto-filled, and how many required manual review. This metrics table shows how effective and safe the inheritance approach is across all metadata fields.

Scope 1 Visuals

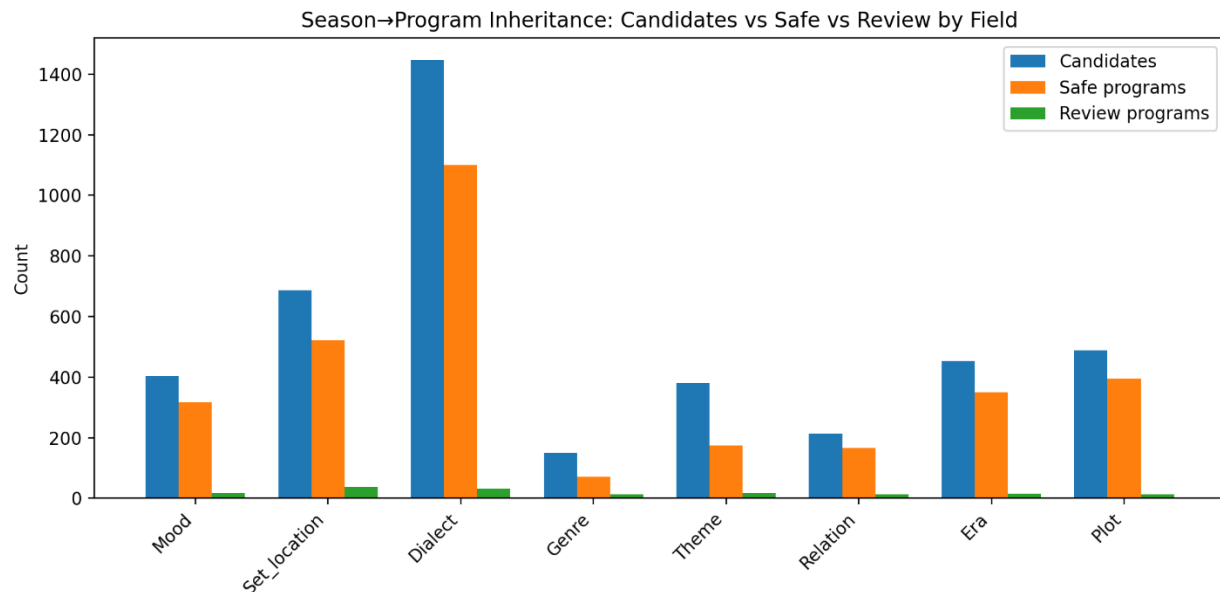
1. Safe inheritance visual (Candidates / Safe / Review)

I started by implementing a safe season→parent inheritance rule for each metadata field (Mood, Genre, Theme, Dialect, Era, Plot, Set_location, Relation). For each field, I focused only on cases where the parent field was missing and at least one season field was populated. I then grouped by Program Id and checked whether all Season values agreed; when they did, the program was labeled **SAFE** and given a proposed parent value, and when seasons conflicted the program was labeled **REVIEW** and left for manual validation.

To summarize this behavior, I created a bar chart that shows, for each field, three counts:

- **Candidates:** program–season pairs where the parent field is missing but the season field is filled.
- **Safe programs:** parent titles where all seasons agree and the field can be safely auto-filled.
- **Review programs:** parent titles where seasons conflict and the field is flagged for manual review.

This visual makes it easy to see for which fields the season data is clean and how much of the missing parent metadata can be recovered safely.

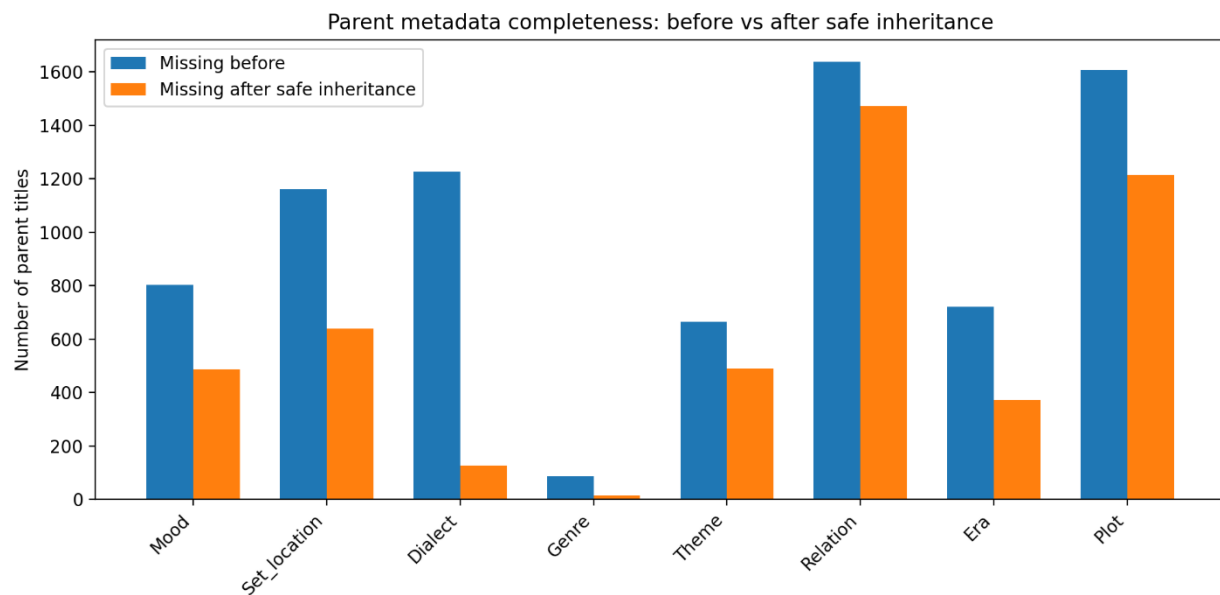


2. Metadata completeness before vs after safe inheritance

To measure the impact of the safe inheritance framework on parent-level metadata completeness, I computed **before vs after** statistics for each field. For every field and every Program Id, I counted:

- **parent_missing_before**: number of parent titles where the Program Count for that field was zero (i.e., the field was missing at the parent level).
- **parent_fixed_by_inheritance**: among those missing parents, how many have at least one SAFE inheritance proposal from seasons.
- **parent_remaining_missing_after**: parents still missing the field after applying all SAFE proposals, i.e., before – fixed.

I then plotted, per field, two bars: “Missing before” vs “Missing after safe inheritance”. For most fields, the number of missing parent values drops by half or more, showing a clear completeness improvement driven purely by safe, rule-based season→parent inheritance, without overwriting any existing parent metadata.



Safe Inheritance Performance Across Metadata Fields

The inheritance metrics show that for most fields, season→parent inheritance is both effective and safe. For example, Mood, Set_location, Dialect, Relation, Era, and Plot all have safe inheritance rates around 0.75–0.80 or higher, meaning that in roughly three out of four candidate cases, all seasons agree and the parent field can be safely auto-filled. In contrast, Genre and Theme have lower safe rates (around 0.48 and 0.46), indicating that season values for these fields are more varied or inconsistent across seasons and therefore require relatively more manual review. Across all fields, the review rate remains low (generally below 0.09), which confirms that the majority of missing parent metadata can be recovered through strictly rule-based inheritance without overriding any existing parent data.

This suggests that fields like Dialect, Plot, Mood, Era, Relation, and Set_location are strong candidates for automated completion, while Genre and Theme should be treated more cautiously and potentially classified as “conditional inheritance” fields that always undergo editorial review before publishing.

Field	Candidates	Safe programs	Review programs	Safe rate	Review rate
Mood	404	316	18	0.78	0.04
Set_location	687	521	37	0.76	0.05
Dialect	1448	1101	32	0.76	0.02
Genre	150	72	13	0.48	0.09
Theme	380	174	17	0.46	0.04
Relation	214	167	14	0.78	0.07
Era	453	350	15	0.77	0.03
Plot	488	395	14	0.81	0.03